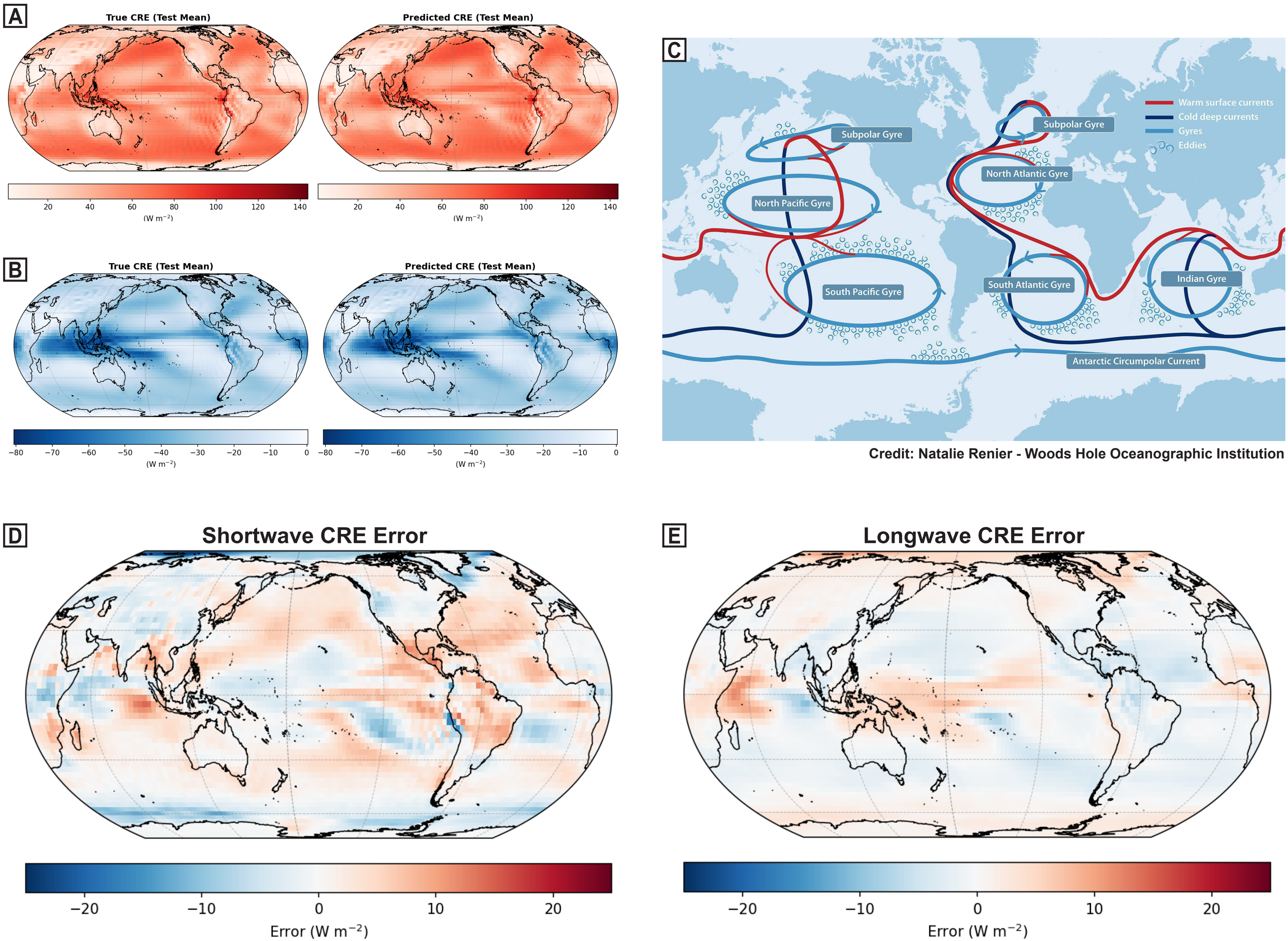
A
True CRE (Test Mean)
Predicted CRE (Test Mean)
20 40 60 80 100 120 140
(W m^{-2})
B
True CRE (Test Mean)
Predicted CRE (Test Mean)
−80 −70 −60 −50 −40 −30 −20 −10 0
(W m^{-2})
C
Warm surface currents
Cold deep currents
Gyres
Eddies
Subpolar Gyre
North Pacific Gyre
South Pacific Gyre
Subpolar Gyre
North Atlantic Gyre
South Atlantic Gyre
Indian Gyre
Antarctic Circumpolar Current
Credit: Natalie Renier - Woods Hole Oceanographic Institution
D
Shortwave CRE Error
E
Longwave CRE Error
−20 −10 0 10 20
Error (W m^{-2})

Figure 1: Model outputs and errors. (A) True shortwave radiation cloud radiative effect (CRE) values, along with model-predicted shortwave radiation CRE values. (B) True longwave radiation CRE values, along with model-predicted CRE values. (C) A diagram depicting the major currents, gyres, and eddies that characterize ocean circulation around the globe. The behavior of these circulation features over time may be a source of error to our model, especially in regions such as the Indian Ocean. (D) Model error for shortwave radiation CRE. The largest errors are located around high latitudes, as well around tropical regions such as Central America and the Indian Ocean. (E) Model error for longwave radation CRE. The model produces less prediction error using longwave radiation than with shortwave radiation, likely because longwave radiation is less dependent on regional processes. However, regions such as the Indian Ocean still exhibit noticeable errors.

Panel (C) was illustrated by Natalie Renier and is sourced from the Woods Hole Oceanographic Institution. The original figure can be found here: https://www.whoi.edu/ocean-learning-hub/multimedia/ocean-circulation-road-map/